MINOR PROJECT / COMPUTER VISION

Real-Time Sign Language Recognition

Pose and hand landmarks with temporal sequence learning • Isolated signs from video and webcam

SOURCE DATASET

3,630 clips / 61 labels

FROM VIDEO TO SIGN LABEL

01 **CAMERA** — Video clip or webcam → 02 **LANDMARKS** — 33 pose + 42 hand points → 03 **SEQUENCE** — 30 frames × 225 features → 04 **LSTM** — Temporal sign classifier → 05 **PREDICTION** — Smoothed label + confidence

PROJECT & DATA

AIM

Recognize one sign at a time from a fixed vocabulary using ordinary camera video.

IMPLEMENTATION

- MediaPipe extracts 33 body landmarks and 21 landmarks per hand; normalized features form 30-frame sequences.
- A two-layer LSTM classifies the sequence. An optional CNN-LSTM model is also implemented.
- The webcam interface applies confidence thresholding and smooths predictions across recent frames.

RESEARCH CONTEXT

The review notes that pose features are compact but may miss facial grammar and fail under occlusion.

3,117 sequences extracted

53 / 61 classes with sequences

Eight source classes currently have no extracted sequences.

RESULTS, LIMITS & NEXT STEPS

SAVED TRAINING PLOT / ACCURACY

~100% **near the final epochs**

in the project's saved split accuracy curve

This is a result shown in the project's saved plot, not an independent benchmark. The code uses a grouped 80/20 split by source recording.

SCOPE & NEXT STEPS

- Recognizes trained labels only; it does not translate continuous sentences.
- Complete missing class extraction; evaluate on unseen signers and varied cameras/lighting.
- Report per-class accuracy and F1, alongside overall accuracy.

Kalinowski, M. & Kostek, B. (2026). Machine learning in sign language: A comprehensive analysis and trend survey. Computer Science Review, 60, 100895. doi:10.1016/j.cosrev.2026.100895.